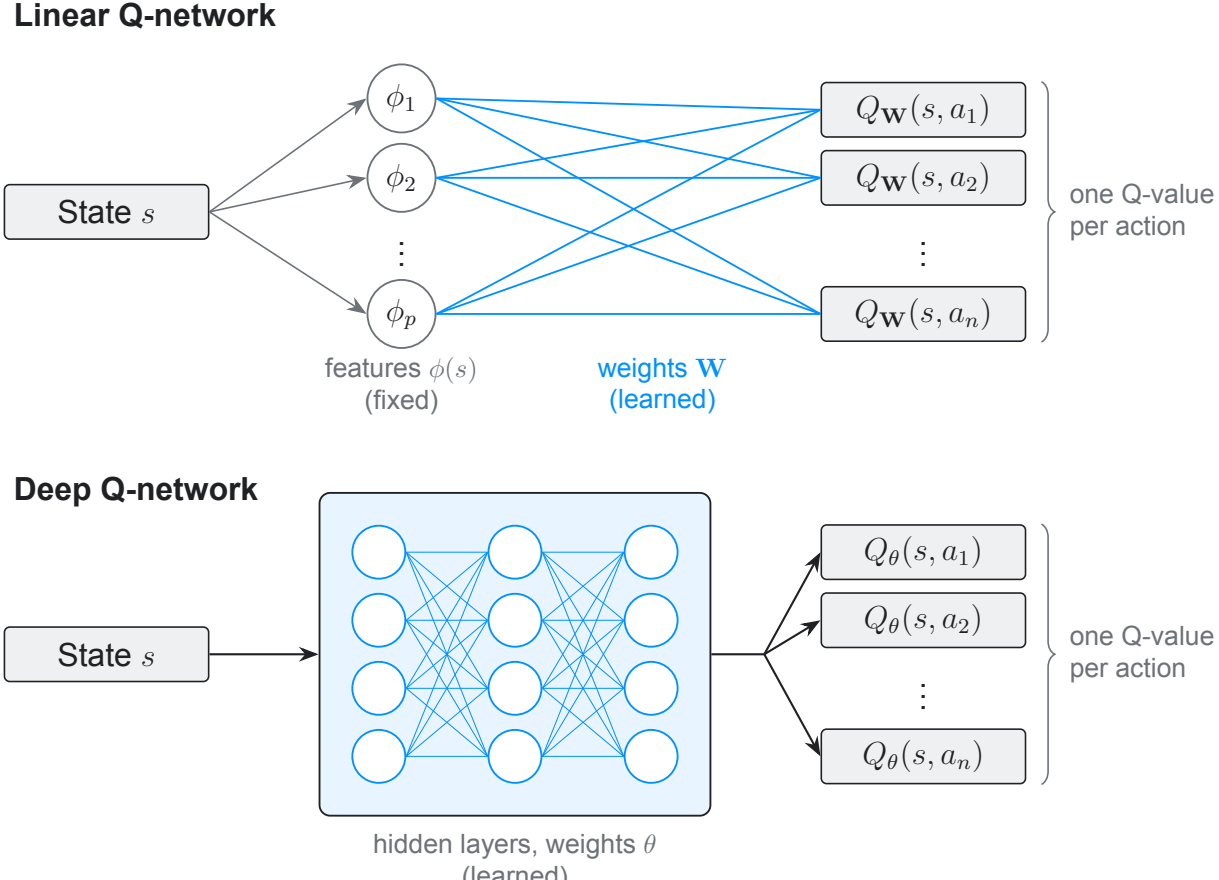
Linear Q-network
State s
ϕ_1
ϕ_2
ϕ_p
features $\phi(s)$
(fixed)
weights $\mathbf{W}$
(learned)
$Q_{\mathbf{W}}(s, a_1)$
$Q_{\mathbf{W}}(s, a_2)$
$Q_{\mathbf{W}}(s, a_n)$
one Q-value
per action
Deep Q-network
State s
hidden layers, weights θ
(learned)
$Q_\theta(s, a_1)$
$Q_\theta(s, a_2)$
$Q_\theta(s, a_n)$
one Q-value
per action